



Tennessee TECH

SAE Aero Design East 2025

Tennessee Technological University

Team 032: Golden Eagle 2

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Statement of Compliance

APPENDIX A - STATEMENT OF COMPLIANCE

Certification of Qualification

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Statement of Compliance

As faculty Adviser:

jt (Initial) I certify that the registered team members are enrolled in collegiate courses.

jt (Initial) I certify that this team has designed and constructed the radio-controlled aircraft in the past nine (9) months with the intention to use this aircraft in the 2025 SAE Aero Design competition, without direct assistance from professional engineers, R/C model experts, and/or related professionals.

jt (Initial) I certify that this year's Design Report has original content written by members of this year's team.

jt (Initial) I certify that all reused content has been properly referenced and is in compliance with the University's plagiarism and reuse policies.

jt (Initial) I certify that the team shall use the Requirements Check & Safety and Airworthiness Inspection checklists to inspect their aircraft before arrival at Technical Inspection and that the team shall submit the completed checklists, signed by the Faculty Advisor or Team Captain, to the inspectors before Technical Inspection begins.

John J. Tester
Signature of Faculty Advisor

3/14/25
Date

Kyle Gandy
Signature of Team Captain

3/16/25
Date

Note: A copy of this statement needs to be included in your Design Report as page 2 (Reference Section 4.3)

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Table 1: Symbols and Acronyms

Symbol	Description
L	Lift
D	Drag
AoA	Angle of Attack
V	Velocity
CL	Coefficient of Lift
CD	Coefficient of Drag
CM	Coefficient of Moment
A	Wing Area
W	Empty Weight
RE	Reynolds Number
Acronym	Description
CAD	Computer Aided Design
FEA/FEM	Finite Element Analysis/Modeling
CAM	Computer Aided Manufacturing
CFD	Computation Fluid Dynamics
GE1	Golden Eagle 1
GE2	Golden Eagle 2
PWM	Pulse Width Modulation
TPU	Thermoplastic Polyurethane
PLA	Polylactic Acid
PETG	Polyethylene Terephthalate Glycol

1.0 Executive Summary

This report details Tennessee Technological University's research, design, testing, and manufacturing process to create our submission for the 2025 Regular Class Competition: The Golden Eagle 2 (GE2). We used the Telemaster 40 RC plane as its design basis because of its ease of manufacturing and assembly. Our craft has a dry weight of 32.87 lbs, with a max payload of 3.43 lbs. This plane is designed to be contrasted with our 2024 craft, the Golden Eagle 1. (GE1)

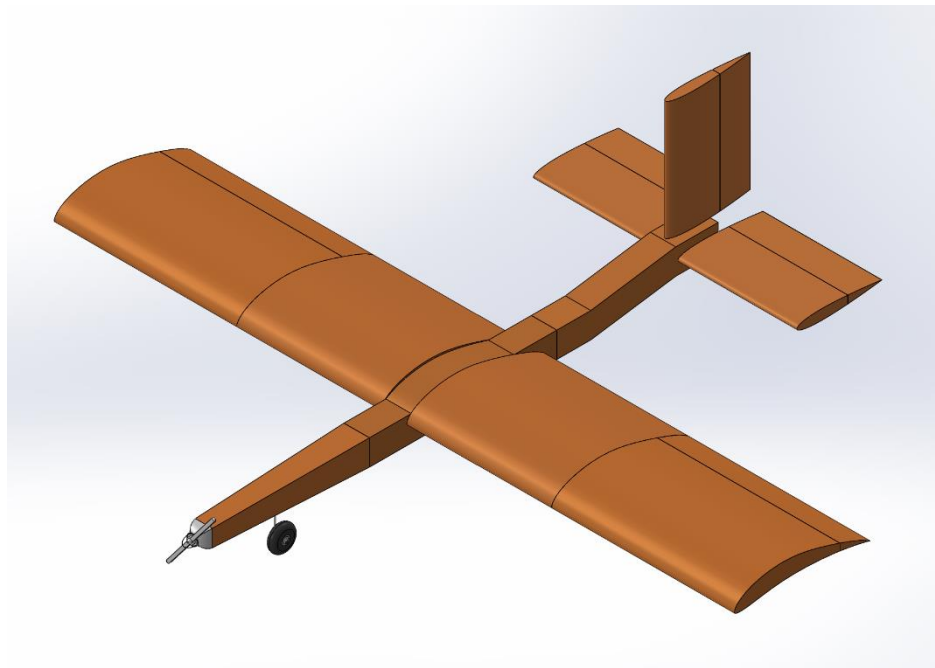


Figure 1: Aerodynamic Model

1.2 Competition Projections and Conclusions

The GE2 is heavy, so our empty weight is the same as our goal weight for this competition. Our takeoff distance at dry weight is calculated as 80.1 ft. The maximum weight our craft can carry is 3.43 lbs, resulting in a takeoff distance of just shy of 100 ft.

1.3 Discriminators

Our design features internal fuselage interfaces with no external hardware, called “snaps”. They use compliant mechanisms that our team designed, 3D printed, iterated, tested, and implemented. We

also emphasized the structural integrity of the airframe, as well as prioritized the ease of manufacturing and assembly of sections to resolve errors we found in the Golden Eagle 1.

2.0 Schedule Summary

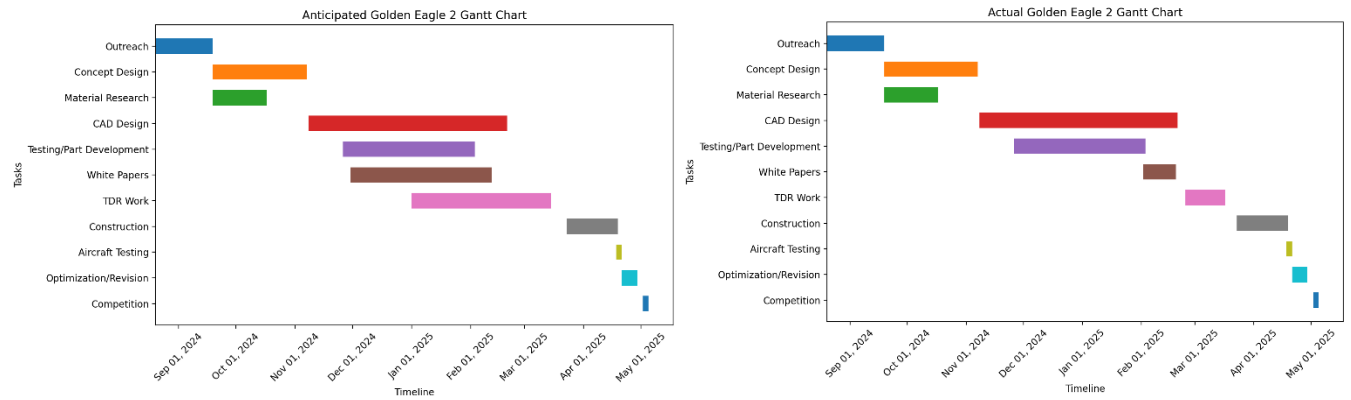


Figure 2: Proposed Schedule (left), Actual Schedule (right)

At the beginning of the 2025 design season, our team established conservative deadlines for when we wanted aspects of our design to be completed. As a second-year team, we expected some of our deadlines to not be met, this discrepancy is reflected in Figure 2. This was the case particularly for the Technical Design Report, as well as specific design decisions. In comparison to the 2024 design season, our time was much more well-structured, and we anticipate a similar improvement next year.

3.0 Table of Referenced Documents, References, and Specifications

Table 2: Table of References

Reference	Description
<i>Airfoil tools</i> . Airfoil Tools. (2025). http://airfoiltools.com/	Aerodynamic performance of airfoils, Cl, Cd
US Department of Commerce, N. (2024, December 16). <i>Climate</i> . National Weather Service. https://www.weather.gov/wrh/Climate?wfo=fwd	Historical weather data for Dallas Love Field
<i>Hitec highlight</i> . RC Groups RSS. (2024). https://www.rcgroups.com/forums/index.php	RC plane construction technique and rules of thumb for design
Hibbeler, R. C. (2022). <i>Mechanics of Materials</i> (11th ed.). Pearson.	Used for beam analysis and deflection calculations
Anderson, J. D. (2011). <i>Fundamentals of aerodynamics</i> (5th ed.). McGraw-Hill.	Aerodynamic Principles/Fundamentals
Sadraey, M. H. (2013). <i>Aircraft design: A systems engineering approach</i> (1st ed.). Wiley.	Fundamentals of aircraft design and system integration

4.0 Design Layout and Trades

Tennessee Tech’s 2024 design season ended prematurely following tolerancing issues due to our manufacturing processes. We reviewed the interfacing systems, layout, manufacturing techniques, and design processes of the GE1 in preparation for this year. When designing the GE2, our main metric was improvement upon last year. We set 2 metrics to meet for each decision:

- Take an existing system and make it as simple as possible.
- Consider the process for how the system will be manufactured.

The Golden Eagle 2’s laser cut wing with no dihedral and rectangular planform directly contrasts the Golden Eagle 1, which had both a split dihedral and taper, and was constructed from foam. This complexity and manufacturing technique more than doubled the time cost of manufacturing. The GE1 also had single-point-of-failure fuselage interfacing sections, and one double shear connection for the

wing-fuselage-landing gear interface. In contrast, the Golden Eagle 2 has 4 independent “snaps” for each fuselage interface. Dual telescoping wing spars pass through 2 reinforced holes in the fuselage. The 4 wing sections are attached with a wing-lock system. The square cross-section of the fuselage is also designed for ease of manufacturing, a design choice we kept from the Golden Eagle 1. We doubled the thickness of the fuselage walls for increased strength and rigidity, re-worked the internal layout, separated the wing, fuselage, and landing gear into 2 interfacing systems, and optimized the cable management in the Golden Eagle 2.

4.1. Overall Design Layout and Size

The Golden Eagle 2 uses the E423 airfoil, with a rectangular wing planform and a wingspan of 160”. The plane length is 133” from nose to tail and the CG is located 54” from the nose of the craft- 3” in front of the Center of Lift. The craft is 47” at its highest point and primarily constructed from ¼" balsa and ¼" aero-lite plywood, with aluminum wing spars and steel landing gear.

4.2 Optimization

To implement the electrical system into the aircraft we first designed the battery bay to have 1/16” balsa sheets to surround it and notches located on the bottom of the aircraft to allow for the wires to pass through. We designed and 3D printed wire clasps to limit movement of wires during flight, making them less likely to disconnect. During disassembly and reassembly to make disconnecting and reconnecting the wires easier, we made a Dupont connector sized notch in the bulkheads that are going to be epoxied in to make

4.2.1 Competitive Scoring and Strategy Analysis

As a second-year design team, our strategy focuses heavily on the Technical Design Report, both for scoring and documentation. We have calculated our max payload to be 3.43 lbs, but our team’s strategy for this year is to overbuild our craft, to have successful competition flight(s), and to create a framework for future teams to reference and build on.

4.2.2 Optimization and Sensitivity Analysis

In analysis of the tail section of the plane, we determined that the placement being directly in line with the main wing may negatively influence the stabilizer due to the slip stream of the wing. Ansys Fluent was used to analyze the turbulence contour with different placements, leading us to raise the tail section 3 inches above the wing to minimize this potential effect. A second simulation confirmed this correction.

4.3 Design Features and Details

Per the 2025 Regular Class ruleset, all subassemblies must be 48” or less along any axis. Our wing team chose to create 4 wing subassemblies, the inner measuring 34”, the outer wing measuring 43”. Our fuselage team guaranteed compliance with this specification by sizing all fuselage sections at a maximum of 44”, our team included this 4” of margin to allow for flexibility if we need to alter aspects of the design.

4.4 Interfaces and Attachments

Our team designed a 3D printed compliant mechanism as the latching mechanism for the fuselage subassembly interfaces. We experimented with PLA, PETG, and TPU as our materials. We ruled out PLA and PETG due to their tendency to fracture instead of elastically deform. TPU is a flexible rubber, so interface failures in TPU components are the result of elastic deformation, which only occurs with loads exceeding 45 lbs. We conducted tensile tests until failure to determine the max force the interface could withstand. Our test results for this are tabulated in Section 7.4, Table 6.

Each interface also has wooden pegs fitting into opposing holes to ensure correct fit and orientation of fuselage subassemblies. This resists against shear and torsional forces acting on the aircraft.

The outboard wing sections use a compliant “snap” mechanism, similar to the fuselage interface connection. This is a technique used in hobby RC aircraft for smaller wing aircraft. The interior wing sections use a commercially available “wing lock” system to mechanically affix the sections together.

5.0 Loads and Environments, Assumptions

5.1 Design Loads Derivations

We approximated our fuselage as shown in Figure 3: fixed at the wing, with the fuselage represented as 2 cantilever beams. Our team calculated the



Figure 3: Fuselage Cantilever Beam Model

beam deflection both under design loads, and shock

loads, which we defined as 6x design load. Our design team is aware of the gross simplifications of these calculations, which is why the Golden Eagle 2 is designed to handle stresses over 10x what our calculations have predicted, to account for any large loads that we may not have anticipated, or errors in calculations. For simplicity, components are treated as point loads acting at their respective centers of gravity. This calculation uses the first and last bulkheads as the beam endpoints, instead of the end of the nose cone and empennage. These calculations are shown in detail in Appendix A.

5.2 Environmental Considerations

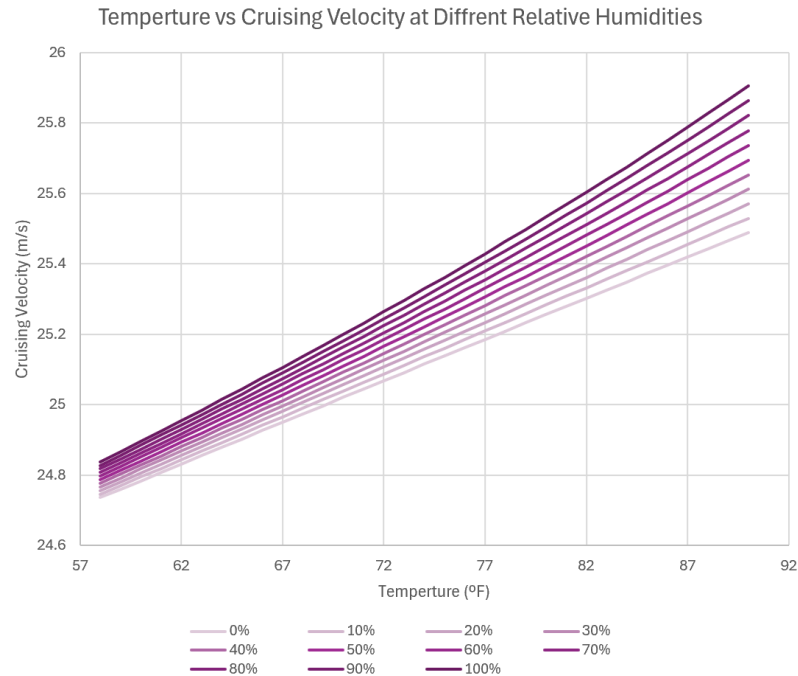


Figure 4: Temperature vs Cruising Velocity at Different Relative Humidities

SAE Aero Design East 2025 is being held in Fort Worth, Texas in early May. We researched the weather conditions for Dallas/Fort Worth using NWS.gov. Our team used historical temperature and barometric pressure readings for May to plot Cruising Velocity as a function of temperature at different humidities. This is shown in Figure . To calculate this, we found the velocity at each temperature and relative humidity using the equation below and computed this in an excel design tool. We expect to have a higher cruising velocity at higher relative humidities and temperatures, causing the GE2 to be traveling at a faster speed during takeoff.

$$v = \sqrt{\frac{2W}{\rho C_L A}}$$

6.0 Analysis

6.1 Analysis Techniques

Our team used SOLIDWORKS for the CAD and FEA of the Golden Eagle 2. We used Ansys Fluent for the CFD of the wing, empennage, and control surfaces of our craft. Design tools were created in Excel to parameterize characteristics of the craft such as lift and drag performance, velocity calculations, lifting surface dimensions, and atmospheric analysis.

6.1.1 Analytical Tools

SOLIDWORKS was used in the design of the whole aircraft after general parameters were found using hand calculations and Excel design tools. Subsequent FEA was performed on some critical components to identify areas where stronger materials or more structural elements were required to be implemented.

For validation of the main wing airfoil, our team used Ansys Fluent to compare the lift and drag coefficients of 3 different airfoils: E423, GOE 4408, and S1223, with the final selection of E423. This airfoil was selected as it had similar performance and aerodynamic efficiency to the optimal S1223 airfoil, but with a simplified geometry for construction purposes.

Ansys Fluent was also used to study the interaction of the wing slip-stream and empennage interaction in 2D, observing velocity contours to determine if the slip-stream would impinge onto the horizontal stabilizer, reducing the aerodynamic performance of the control surfaces, and creating the possibility of a stall. The horizontal stabilizer was then positioned 3 inches higher than the main wing when level on ground to mitigate potential stall from the slip stream due to the main wing.

6.2 Performance Analysis

6.2.1 Runway/Launch/Landing Performance

Using the data in section 7.2, the team calculated the takeoff distance for additional payload vs. takeoff distance, displayed in Figure 5. Summing the forces the plane experiences in the direction of motion yields the following equation:

$$\Sigma F = m \cdot a = T(t) - D - F_r$$

This is the case when $T(t)$ is the function of thrust. D represents total drag on the aircraft, and F_r is rolling friction experienced by the wheels. Total mass of the plane is depicted as m , and acceleration is a . Expanding these parameters and solving for acceleration, we receive the following equation:

$$a = \frac{T(t)}{m} - \frac{C_d \rho A}{2m} - \mu_r g$$

Some important assumptions include an environment without wind. Additionally, air density, ρ , is assumed to be 0.0023 slug/ft³. Coefficient of rolling friction, μ_r , is assumed to be 0.05. Finally, drag coefficient, C_d , is assumed to be 0.15. These values were chosen as approximations, leaning towards the worst case of performance to account for unknown or unforeseen inefficiencies in the analysis. Figure 5 is derived, starting from 0 ft/s, displaying takeoff distance as a function of payload.

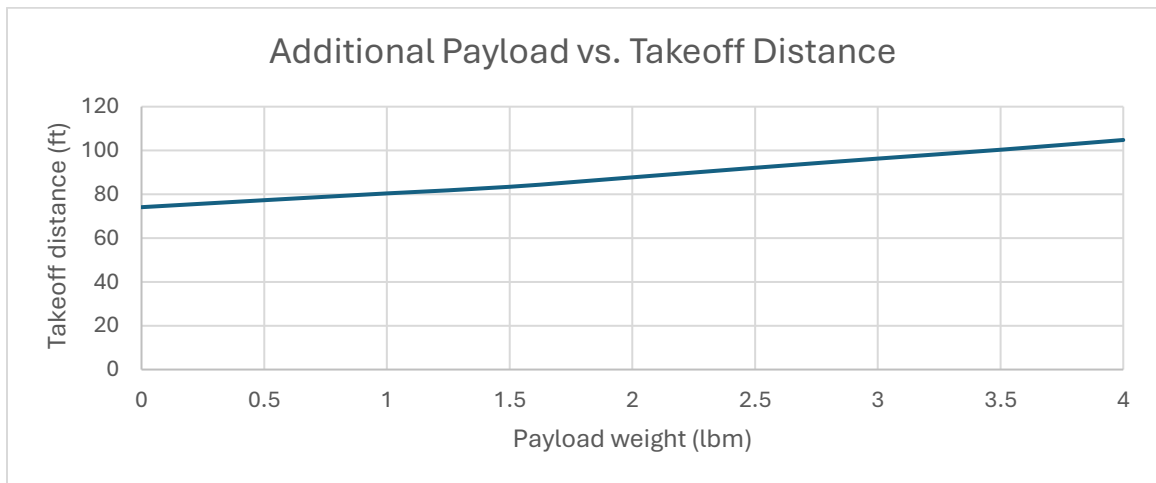


Figure 5: Additional Payload vs Takeoff Distance

While analyzing landing performance, we used the same equation outlined in the takeoff performance analysis above, using cruising velocity and $T(t) = 0$ lbf as initial condition. This process yielded Figure 6 below.

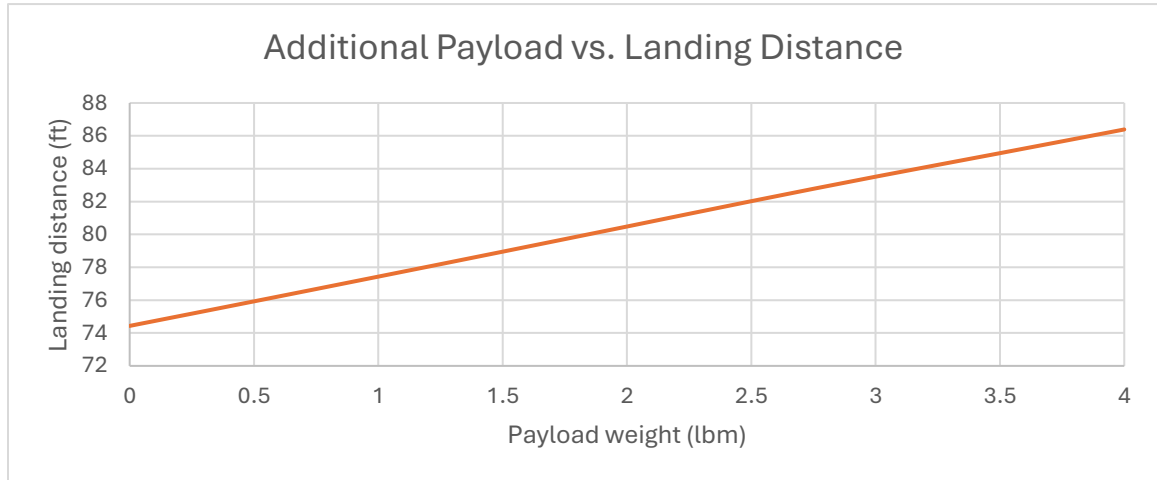


Figure 6: Additional Payload vs Landing Distance

6.2.2 Flight and Maneuver Performance

The design team elected to use the NACA 0012 airfoil in our empennage. It is a symmetric airfoil chosen to minimize both the C_d and C_l , with a 20" chord for easy servo integration. Due to the low airspeed of the craft, our ailerons are 43" span with a 7.16" chord, as we anticipate our craft to be stable in the roll axis. Our empennage control surfaces are sized using average values for glider type aircraft. These values are in the middle of the recommended ratio for control surface size compared to wing area. This ensures controllability in the pitch and yaw axes. Since the aircraft is flying at relatively low speeds, these control surfaces will provide enough maneuverability in relation to the overall speed.

6.2.3 Shading/Downwash

We used Ansys Fluent to analyze the slip stream of the main wing and its effect on the horizontal stabilizer. Using 2D simulation with the specified chord lengths of each component at flow velocity of 30 ft/s, we can see that the slip stream does not interfere with the empennage in level flight. The height

of the empennage was adjusted based on simulation with both components at the same height from the ground. This configuration caused the slip stream to be much closer to the horizontal stabilizer, causing a potential stall at a higher AOA of the plane. Figure 7 shows the turbulence contour of the wing and horizontal stabilizer at 30 ft/s.

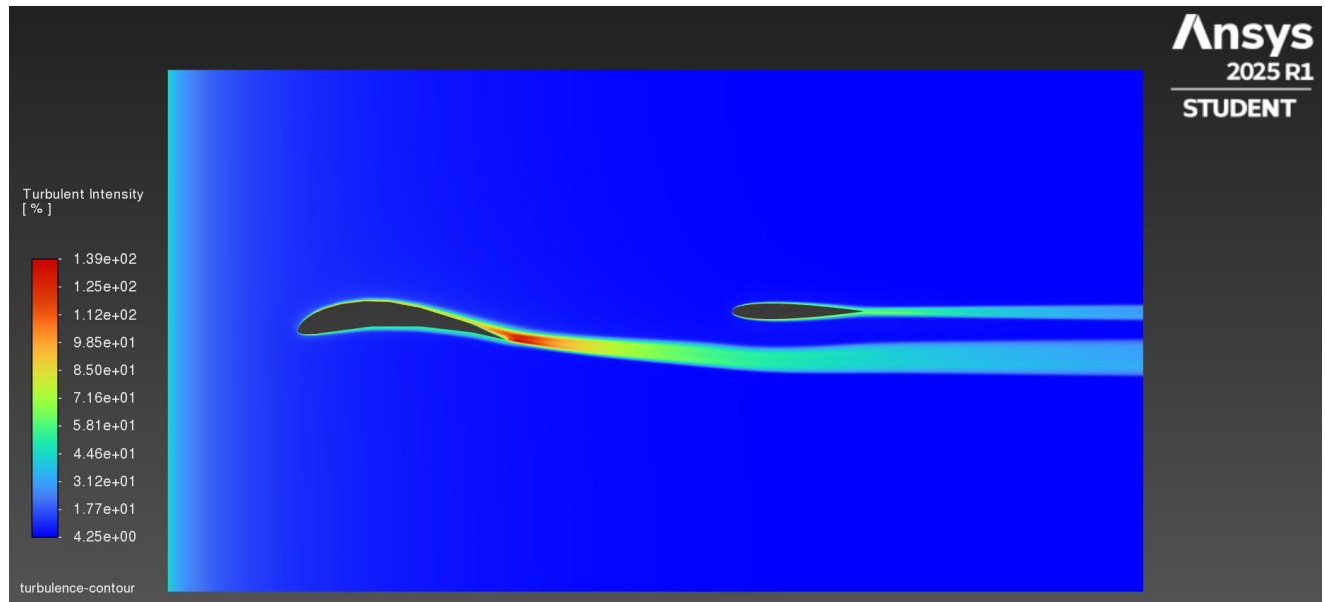


Figure 7: Turbulence Contour of Wing and Horizontal Stabilizer

6.2.4 Dynamic and Static Stability

The longitudinal static stability of the plane is ensured by placing the center of gravity ahead of the center of lift. This is done to encourage the plane to nose down when a potential stall scenario takes place, allowing the plane to more easily recover and continue flying. Using a static margin calculation based on CG and center of lift, the static margin of the loaded and unloaded plane will be 0.0938.

Some static stability in the roll axis will be from the wing deflection, causing a slight dihedral of the wing. This will allow the plane to contribute some lift force in the direction of the deflection, allowing a small amount of instability in the roll direction to self-correct.

Yaw stability will come from pilot input and trim on the rudder. This axis is predicted to be stable as the plane will be symmetrical when assembled, having little to no unbalanced forces in ideal flight in this axis.

6.2.5 Aeroelasticity

The Golden Eagle 2 was built with an emphasis on structural rigidity. This higher structural stiffness results in a higher harmonic frequency of the structure. As a result, oscillatory effects from aerodynamic forces will not grow exponentially. This minimizes aeroelastic flutter in the structure. Deflection of any component will be minimal due to this focus on structural integrity.

6.2.6 Lifting Performance, Payload Prediction, and Margin

Based on previous years' weather data, we assume the temperature at the competition will be between 58°F and 90°F. Using previous weather data, an average air density of $\rho = .00238 \text{ slug/ft}^3$ was estimated. The takeoff distance for various payload weights was calculated, similar to the method found in Section 6.2.1, using an average air density computed from historical weather data at the competition site. The payload weight corresponding to a takeoff distance of 100 ft was calculated as 3.43 lbs. The value of payload weight corresponding to a takeoff distance of 80 ft.- allowing 20 ft. of margin in takeoff distance to account for the assumptions used in our calculations- is 0lbs. Thus, the dry weight of the Golden Eagle 2 is the max design weight. Any payload added cannot guarantee takeoff in under 100 ft. due to non-idealized conditions.

6.3 Structural Analysis

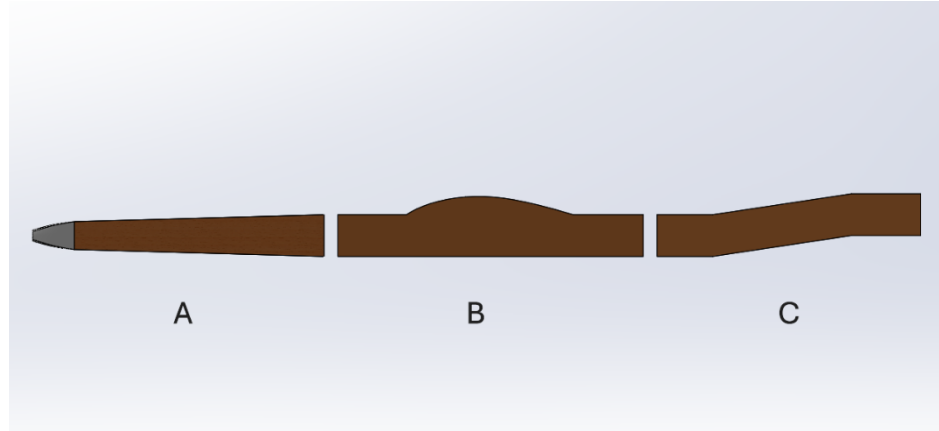


Figure 8: Fuselage Section Naming Convention

The airframe of the fuselage of the Golden Eagle 2 is a minimum of $\frac{1}{4}$ " thick at any point. The fuselage of the craft is composed of 3 subassemblies- called "sections". Figure 8 shows the fuselage section naming convention. Section B, containing both the wing and rear landing gear, and thus undergoing the most significant structural loads, is constructed from $\frac{1}{4}$ " aero-lite birch plywood. The nose and empennage sections are constructed from $\frac{1}{4}$ " balsa because the stresses present in those sections are less significant. The GE2 is designed to withstand forces more than 5x those we anticipate. This is reflected dramatically in the weight of the craft. As a second-year team, our systems to analyze structural analysis have not been fully established, so our team opted to build the Golden Eagle 2 to withstand such forces, eliminating concerns about the structural integrity of the airframe, despite lacking developed stress models.

6.3.1 Applied Loads and Critical Margins Discussion

The most significant loads are present in section B, where the rear landing gear, the main wing, and the payload bay are located. This section of the craft experiences the highest stresses during takeoff and landing, as well as during turning flight. For this reason, at the spar interface, we have doubled the thickness of the plywood and made this interface double sheared.

Section A contains both the main and receiver batteries, as well as the motor and nose gear. This section experiences less stress during takeoff and landing and thus can be constructed out of ¼” balsa. To reinforce the nose gear of the craft, as well as other regions experiencing heightened loads, we have implemented ¼” plywood bulkheads spanning the fuselage cross-section to distribute the landing forces across a greater surface area.

Section C supports the empennage of the craft, so its primary purpose is as a moment arm counteracting the weight of the empennage. For our calculations, we represented the empennage and its associated electronics as a 3.3 lb point load acting at the center of the empennage/fuselage interface.

6.3.2 Mass Properties and Balance

The center of gravity for the Golden Eagle 2 is located 49.13” from the datum of the aircraft, and 3” in front of the center of lift. In the event of a stall, this design will reduce the angle of attack and assist stall recovery. Before finalizing the CG location during design, we kept the length of one fuselage section behind the wing variable, as well as the location of the battery and payload bays, to give us flexibility in adjusting the center of gravity. The triangle footprint of the landing gear defines the bounds that the center of gravity must remain within to prevent tipping. At rest, the CG is minimum 6” from any boundary. This results in increased stability during take-off and landing.

7.0 Assembly and Subassembly, Test and Integration

7.1 Maximum Thrust Data

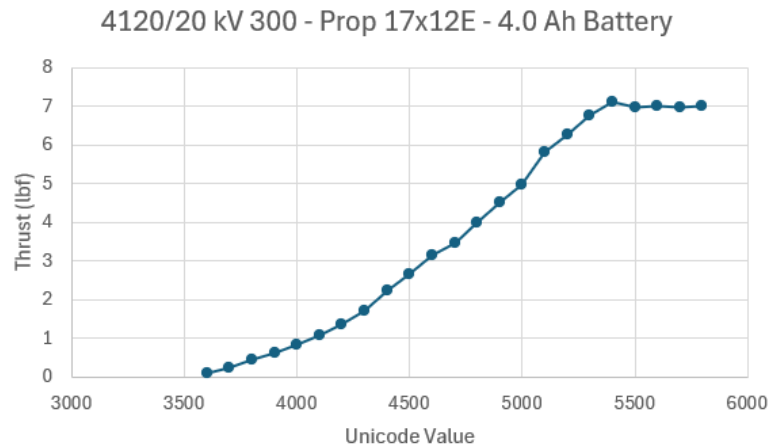


Figure 9: Thrust profile of 17x8E Propeller on 4120/20 Kv 465 Cobra motor

The electrical team designed and built a test stand to measure thrust output and runtime from combinations of batteries, propellers, and motors. In the first test stand iteration, the thrust data collected by a strain gauge sensor was inaccurate compared to measuring the same setup with a scale. Pivoting to a less automated data collection method, the maximum thrust for various propellers is shown in Table 3. Using this data the Cobra 4120/20 motor with a 300 Kv rating paired with a 17x12E propeller provides 7.6 lbf, the maximum thrust out of the propellers the team tested. Thrust data taken at increasing throttle values is shown above in Figure 9.

Table 3: Maximum Thrust values from Motor and Propeller Pairings

<u>All pairings tested with 4.0 Ah battery</u>	14x7E Propeller	16x8E Propeller	17x12E Propeller
Cobra 4120/20 Kv300 Motor	6.3875lbf	7.0125 lbf	7.6875 lbf

To gather thrust data for motor and propeller combinations (Figure 9), the microcontroller controls the motor by replicating PWM values sent by a receiver with equivalent Unicode16 values. This allowed for the motor and propeller combinations to be tested in near identical conditions. The thrust

data required using a cellphone to record values from a scale positioned at the end of the test stand for a 1:1 mechanical advantage as shown in Appendix A.

7.2 Maximum Runtime

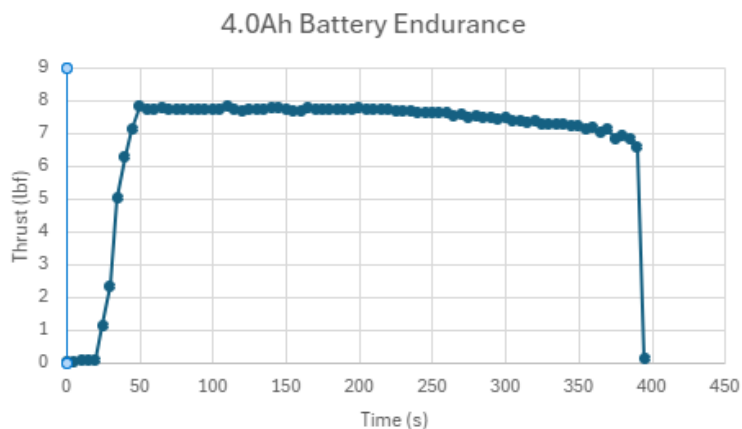


Figure 10: Runtime graph for 4.0 Ah battery chosen as the battery for the competition flight

The run time data followed the same procedure as section 7.1, replacing the microcontroller with a receiver and transmitter. The transmitter is set to max throttle and records runtime for the entire flight duration. All battery runtimes tested are shown in Table 4. Taking into consideration the quick flight times from spectating previous competitions, we have chosen a 4.0 Ah battery by Turing for its longer runtime and relatively lower weight compared to the other batteries tested. The graph showing the runtime for the 4.0 Ah battery selected by the team is shown in Figure 10.

Table 4: Runtimes for multiple batteries at full throttle

	3.3 Ah Battery (Weight: 1.096 lbs)	4.0 Ah Battery (Weight: 1.368 lbs)	4.5 Ah Battery (Weight: 1.653 lbs)	5.2 Ah Battery (Weight: 1.614 lbs)
Runtime (s)	360 s	390 s	365 s	517 s

7.3 Calibration Curves

The team created density calibration curves for both balsa and plywood by weighing our stock. We imported these values into SOLIDWORKS for the most accurate weight prediction during our mass-balance analysis of the craft. We also used these density values during our beam bending analysis to estimate the Young's Modulus of the balsa, due to extreme variability in this triangular property. These values are tabulated in Table 5.

Table 5: Calculated Density of Materials

Density (g/in ³)						
	Plywood			Balsa		
Dims (in)	0.25x6x48	0.25x12x48		0.125x6x48	0.25x6x48	0.25x3x48
Density (g/in ³)	6.735	6.713		1.854	2.039	1.334
Density (lb/in ³)	0.01485	0.01480		0.00409	0.00450	0.00294

7.4 Fuselage Interface Testing

Table 6: Fuselage Interface Clip Test Data

TPU										
Max force in (lbf)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8	Trial 9	Trial Average
100% Wall	49.9	39.2	30.5	34.9	33.0	32.1	28.3	29.2	26.6	33.74
100% Infill	98.8	82.8	61.6	56.2	35.8	50.9	53.5	50.4	48.5	59.83
4 Wall 75% Infill	42.4	32.3	38.0	38.8	24.2	32.9	31.4	35.0	28.2	33.69

Fuselage designed and built a cost-effective test stand to measure the tensile strength of the snaps used for the fuselage subassembly interfaces. We fixed one 4" test section and applied a constant, increasing force to the opposing section via a ratchet strap. It uses a 110 lb digital force gauge to measure the peak force the interface withstood before failure. The results of this testing are summarized in Table 6. 100% infill solid TPU is the best option, averaging 59.83 lbs of tensile force. We speculate the increase in strength in the 100% wall snap is due to the print orientation. The texture of the infill surface

has ridges when compared to smooth 100% wall snaps, increasing the coefficient of friction between the snap and the bulkhead.

8.0 Manufacturing

Critical to our success this year is the creation of 2 roles in our organization: Design Lead and Manufacturing Lead. They ensure both design and manufacturing are considered at all points in the process. After the issues experienced in the GE1, the Golden Eagle 2 prioritizes ease of manufacturing. The rectangular fuselage cross-section allows for a flat surface during assembly. Tabs, slots, holes, and pegs are all employed to orient parts relative to one another. Laser cutting allows for part information to be imported from CAD directly into CAM software, minimizing processes requiring human precision, thereby reducing total error. Using the same materials across sub-teams allows for bulk purchasing of stock and more accurate density values from our calibration curves.

During assembly of the fuselage, components will be fixed to metal tables using magnetic fixtures to align pieces, which will then be bonded together using epoxy. The wing, vertical stabilizer, and horizontal stabilizer will be constructed in the same fashion, using sliding calipers to ensure uniform spacing between ribs. Once all the components for each section are dry fit, the epoxy will be applied and allowed to cure. Due to the laser cutter's inherent precision, the wooden parts of each section will sufficiently fit together with minimal sanding. Combining accurate laser cut parts with commercially available components with specified tolerances means that the construction of each component will self-align when dry fit, ensuring dimensional accuracy when compared to the CAD model.

9.0 Conclusion

The primary goal of the Golden Eagle 2 is to maximize simplicity of design and manufacturing, while extensively documenting the aircraft. This design projects a maximum gross weight of 32.87 lbs, for a maximum payload of 3.43 lbs . The max runtime of the GE2 is 6 minutes and 10 seconds at a cruising velocity of 26.04 ft/s. We project the craft will take off in 80 feet at dry weight and just shy of 100 feet at max weight. As a university in the early years of developing its Aerospace Engineering program, our team is creating the research for future design teams to reference, building the repository of design tools, information, and resources that many other teams readily have. Our goal for this competition year is a well-structured design that the next team will improve upon.

Appendix A

Golden Eagle 2 Design Loads:

Table of Variables

E : modulus of elasticity (psi)

l : moment arm w.r.t. wing (in)

δ : deflection (in)

I : moment of inertia (in⁴)

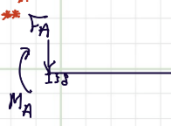
Assumptions and Simplifications

- The fuselage can be modeled as 2 cantilever beams, treating the 2" OD main wing spar as fixed.
- Weight of section B is treated as plywood to correctly account for self-weight
- Assume all loads act vertically
- Assume all loads are point loads
- Assume E_{ply} for entire beam
- Assume homogeneous material

Fuselage Bending Analysis:

$$\delta = -\frac{F_A l_A^3}{3EI}$$

*** diagram mirrored for section WE, F_A becomes F_E .



$$I = \frac{BD^3 - bd^3}{12} = 380.94 \text{ in}^4$$



$35 \times 10^6 < E_{ply} < 55 \times 10^6$
 *units psi. Has high variability due to material inconsistencies
 $E = 48 \times 10^6$ psi: used in design loads analysis
 $E = 35 \times 10^6$ psi used in shock loads analysis

deflection at design load A: $\delta_A = \frac{(6059 \text{ lbs})(60 \text{ in})^3}{3(48 \times 10^6 \text{ psi})(380.94 \text{ in}^4)} = -0.002386 \text{ in}$

$F_{A, \text{shock}} = 36354 \text{ lbs}$

$\delta_A = -\frac{(36354 \text{ lbs})(60 \text{ in})^3}{3(35 \times 10^6 \text{ psi})(380.94 \text{ in}^4)}$

deflection at design load E: $\delta_E = \frac{(5757 \text{ lbs})(64 \text{ in})^3}{3(48 \times 10^6 \text{ psi})(380.94 \text{ in}^4)} = -0.002751 \text{ in}$

deflection at shock loads (ex design loads)

$F_{E, \text{shock}} = 34543 \text{ lbs}$

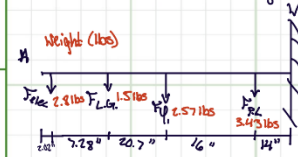
$\delta_{A, \text{shock}} = -0.01431 \text{ in}$

$\delta_E = \frac{(34543 \text{ lbs})(64 \text{ in})^3}{3(35 \times 10^6 \text{ psi})(380.94 \text{ in}^4)}$

$\delta_{E, \text{shock}} = -0.01651$

Design Load Derivations:

Section AW (Front-Wing)

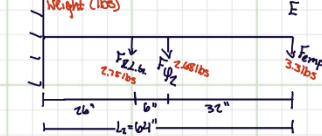


$$\sum M_W = M_A = [F_{\text{Fuselage}} \cdot l_{\text{Fuselage}}] + [F_{L.A.} \cdot l_{L.A.}] + [F_F \cdot l_F] + [F_{RL} \cdot l_{RL}]$$

$M_A = 363.514 \text{ in-lbs}$

$F_A = \frac{M_A}{l} = \frac{363.514 \text{ in-lbs}}{60 \text{ in}} = 6059 \text{ lbs}$

Section WE (Wing-empennage)



$$\sum M_W = [F_{L.A.} \cdot l_{L.A.}] + [F_F \cdot l_F] + [F_{Emp} \cdot l_{Emp}] = M_E$$

$M_E = 368.46 \text{ in-lbs}$

$F_E = \frac{M_E}{l_E} = 5.757 \text{ lbs}$



Figure 11: Picture of unpowered test stand with safety cage removed

Technical Data Sheet

Team Name: Tennessee Tech SAE Aero Design

Team Number: 032

School: Tennessee Technological University

The following is the payload prediction curve we generated. The steps taken to generate this graph and its outcomes are detailed in section 6.2.6.

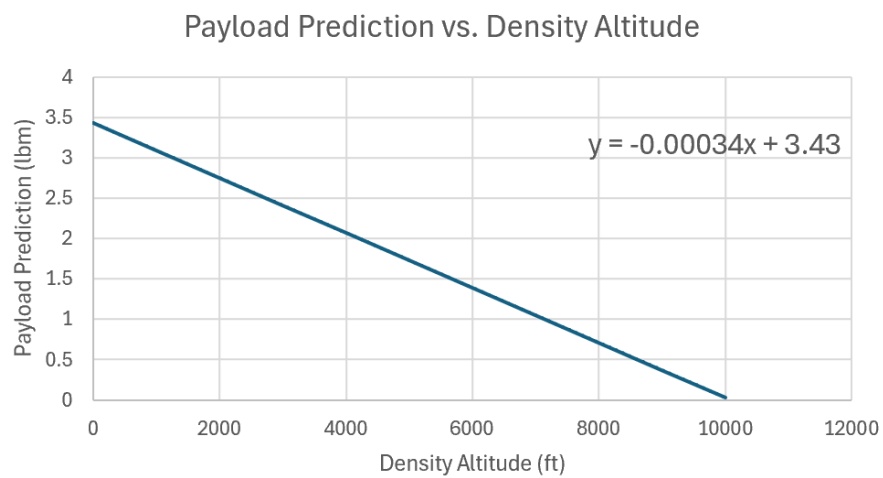


Figure 12: Payload Prediction vs Density Altitude

